Does Auxiliary Spatial Supervision Make Vision-Language-Action Models Safer? A Three-Axis Mechanism Analysis

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Abstract

Adding auxiliary spatial prediction heads (e.g., occupancy, depth) to Vision-Language-Action (VLA) models during fine-tuning is a natural strategy for improving collision avoidance. We conduct a systematic mechanism analysis on a 7.6B-parameter VLA using a six-condition factorial ablation on the LIBERO-Spatial benchmark, disentangling *what* drives safety outcomes along three axes: (1) **gradient flow existence**—severing the backward gradient from an auxiliary head to the vision encoder (Detach) degrades collision avoidance by -3.4 pp relative to a no-auxiliary baseline, while all gradient-flow conditions remain within seed noise of the baseline; (2) **content irrelevance**—replacing ground-truth BEV occupancy targets with spatially-random targets (Random OCC, CAR $= 0.789$) or egocentric depth (Depth, CAR $= 0.780$) yields performance statistically indistinguishable from full occupancy supervision (Full OCC, CAR $= 0.797$); (3) **temporal sufficiency**—a dose-response analysis with cutoff ablations at 15K/25K/35K/45K steps reveals that early-phase gradient flow exposure (first 70% of training) suffices for full protection, while a reverse-cutoff experiment shows that late-phase exposure alone fails (CAR $= 0.752$ vs. 0.785 for early exposure). Gradient norm tracking confirms that auxiliary gradients decay to $<5\%$ of action gradients by the saturation point, explaining the lossless removal. All findings replicate on LIBERO-Goal with identical condition ordering. Our results establish that auxiliary gradient flow is a *necessary condition* for preventing safety degradation during VLA fine-tuning, but the spatial content of supervision is secondary to its mere presence.

1 Introduction

Vision-Language-Action (VLA) models trained end-to-end on large-scale data have emerged as a dominant paradigm for robotic manipulation [1, 2]. A recurring concern in deploying these models is *safety*: VLA policies can succeed at tasks while incurring frequent collisions with obstacles and the environment. A natural hypothesis is that equipping VLA models with auxiliary spatial prediction tasks—such as BEV occupancy prediction or depth estimation—during fine-tuning should improve 3D spatial awareness and therefore reduce collisions.

In this work, we subject this hypothesis to a rigorous mechanism analysis. Rather than simply comparing "with auxiliary head" versus "without," we design a factorial ablation that independently varies three factors: (i) whether the auxiliary head's gradients flow back into the vision encoder, (ii) what spatial content the auxiliary targets encode, and (iii) when during training the gradient signal is active. This three-axis decomposition reveals a surprising picture.

Core finding. Auxiliary gradient flow is a *necessary condition* for preventing safety degradation—but the spatial content of the auxiliary targets is largely irrelevant. When gradient flow is severed (the Detach condition), collision avoidance rate (CAR) drops by 3.4 pp below a vanilla baseline that uses no auxiliary head at all. In contrast, all conditions that preserve gradient flow—whether predicting ground-truth BEV occupancy, random spatial noise, or egocentric depth—perform comparably to the vanilla baseline. The mechanism is not "learning 3D structure helps avoid collisions" but rather "auxiliary gradient flow provides lightweight regularization that prevents encoder degradation during fine-tuning."

We further show that this regularization effect is *temporally front-loaded*: gradient flow during the first 70% of training suffices for full protection, and late-phase exposure alone cannot compensate for early absence. A dose-response analysis with five cutoff points traces a saturation curve that aligns with gradient norm measurements, providing a mechanistic account of the temporal dynamics.

Contributions.

1. A three-axis mechanism analysis establishing that auxiliary gradient flow existence, not spatial content, is the necessary condition for preventing VLA safety degradation during fine-tuning.

2. A temporal dose-response analysis with forward and reverse cutoff experiments that identifies the critical window for gradient flow exposure and establishes a causal direction (early $\gg$ late).

3. Cross-benchmark replication on LIBERO-Goal confirming that all six conditions preserve their relative ordering, eliminating single-benchmark bias.

2 Related Work

VLA Models and Safety. Recent VLA architectures [1, 2] fine-tune large vision-language backbones for robotic control. Safety is typically treated as a deployment concern addressed via control-theoretic methods (e.g., Control Barrier Functions), external monitoring, or constrained optimization [3]. Few works examine how the fine-tuning procedure itself affects collision behavior.

Auxiliary Tasks in Robot Learning. Auxiliary prediction tasks have a long history in reinforcement learning and robot learning [4, 5]. Prior work typically evaluates auxiliary tasks by their effect on task performance (success rate), not on safety metrics such as collision avoidance. Our factorial design isolates the mechanism by which auxiliary tasks affect safety.

Gradient Flow and Regularization. The role of auxiliary gradients as implicit regularizers has been studied in multi-task learning [6, 7]. Our finding that gradient flow presence matters more than target content connects to work on random-target training [8], which shows that even nonsensical supervision can shape representations through gradient dynamics.

3 Method

3.1 VLA Architecture and Auxiliary Heads

We study a 7.6B-parameter VLA model comprising dual vision encoders (DINOv2 and SigLIP), a language model backbone, and an action prediction head. Fine-tuning applies LoRA (rank = 32, $\alpha = 16$) to both vision encoders.

Occupancy Head. The auxiliary occupancy head takes concatenated features from both vision encoders ($d = 2176$) and predicts a 64×64 BEV occupancy grid via two transposed convolution layers. The head contributes 1.2M parameters (0.016% of the full model). Occupancy loss uses Focal Loss ($\alpha = 0.25$, $\gamma = 2.0$) with an exponentially decaying weight $w(t) = 0.01 \cdot \exp(-t/50000)$.

Depth Head. An alternative auxiliary head predicts egocentric depth maps at 64×64 resolution from the same 2176-d concatenated features, using Smooth-L1 loss with identical weighting schedule. Architecture and parameter count are matched to the occupancy head.

3.2 Experimental Design: Six-Condition Factorial Ablation

We design six training conditions that independently vary gradient flow, target content, and temporal schedule:

1. **Vanilla**: No auxiliary head. Pure action loss baseline.
2. **Full OCC**: Occupancy head with ground-truth BEV targets. Full gradient flow to encoders throughout training.
3. **Random OCC**: Same architecture as Full OCC, but targets are spatially random binary grids (resampled each step). Gradient flow preserved; spatial content removed.
4. **Depth**: Depth head with ground-truth egocentric depth targets. Tests whether egocentric spatial structure outperforms allocentric.
5. **Detach**: Occupancy head with ground-truth targets, but `.detach()` severs the gradient path from the occupancy loss to the vision encoders. The head trains (consuming compute) but cannot regularize the encoder.
6. **Cutoff@t**: Occupancy head active for steps $[0, t]$, then removed. Tested at $t \in \{15\text{K}, 25\text{K}, 35\text{K}, 45\text{K}\}$.

Additionally, a **Reverse Cutoff@35K** condition activates the occupancy head only during steps $[35\text{K}, 50\text{K}]$, forming a temporal mirror of Cutoff@35K.

3.3 Metrics

Task Success Rate (TSR): fraction of episodes where the task goal is achieved. **Collision Avoidance Rate (CAR)**: fraction of episodes completed without any collision with obstacles. CAR is the primary safety metric. All evaluations use 500 episodes (50 rollouts × 10 tasks) with a fixed evaluation seed.

4 Experiments

4.1 Setup

All experiments use the LIBERO-Spatial benchmark (10 tabletop manipulation tasks) as the primary testbed, with LIBERO-Goal as a replication benchmark. Training runs for 50K steps on 4× RTX PRO 6000 GPUs (∼12.9h per run at 1.08 it/s). Each main condition is evaluated with 3 seeds (seeds 42–44); Depth uses 4 seeds. Cutoff and reverse-cutoff experiments use 1 seed each. The learning rate is 5×10^{-4} (constant schedule).

4.2 Axis 1: Gradient Flow Existence

Table 1 presents the six-condition comparison on LIBERO-Spatial.

The dominant pattern is a binary split. All gradient-flow conditions (Full OCC, Random OCC, Cutoff@35K, Depth) cluster within ±1.7 pp of Vanilla—well within the seed standard deviation ($\sigma \approx 0.012$–0.037). The Detach condition, which severs gradient flow while maintaining identical forward-pass computation, falls 3.4 pp below Vanilla.

The Detach condition also exhibits extreme TSR instability: $\sigma_{\text{TSR}} = 0.163$ (one seed collapses to TSR = 0.624), compared with $\sigma_{\text{TSR}} < 0.050$ for all other conditions. This

Table 1: Six-condition comparison on LIBERO-Spatial (3 seeds each; Depth uses 4 seeds). CAR is the primary safety metric. Δ CAR is relative to Vanilla.

Condition	Seeds	CAR	σ_{CAR}	Δ
Full OCC	3	0.797	0.032	+1.7
Random OCC	3	0.789	0.031	+0.9
Cutoff@35K	3	0.785	0.015	+0.5
Vanilla	3	0.780	0.012	—
Depth	4	0.780	0.037	$\pm$0.0
Detach	3	0.746	0.002	$-$3.4

suggests that the absence of encoder regularization not only harms safety but destabilizes task execution.

Interpretation. Gradient flow from the auxiliary head is a *necessary condition* for avoiding safety degradation. Without it, the auxiliary head imposes computational overhead (forward pass, loss computation) without any regularization benefit, actively harming the model.

4.3 Axis 2: Spatial Content Irrelevance

Among the gradient-flow conditions, the content of auxiliary targets has negligible effect:

- Full OCC (ground-truth BEV): CAR = 0.797
- Random OCC (random binary grids): CAR = 0.789
- Depth (ground-truth egocentric depth): CAR = 0.780

The spread across these three conditions is 1.7 pp, comparable to within-condition seed variance ($\sigma \approx 0.031$–0.037). In particular, Depth matches Vanilla exactly (CAR = 0.780 with 4 seeds), confirming that even geometrically meaningful egocentric supervision provides no advantage over the no-auxiliary baseline.

This rules out the intuitive hypothesis that "better spatial targets yield safer policies." The mechanism is content-agnostic: any auxiliary head that preserves gradient flow provides equivalent regularization.

4.4 Axis 3: Temporal Sufficiency

4.4.1 Dose-Response Analysis

We train four additional cutoff conditions, each receiving occupancy gradient flow for a prefix of training before removal. Table 2 presents the complete dose-response curve.

The dose-response curve exhibits a classic saturation shape. The marginal CAR gain per 10K additional steps of gradient flow (Table 3) peaks at 1.6 pp/10K in the 15K–25K interval and declines to 0.6 pp/10K beyond 45K. The Cutoff@35K condition (70% of training) already matches Vanilla, indicating that the remaining 30% of gradient flow is unnecessary.

Table 2: Dose-response: CAR as a function of gradient flow duration. Cutoff@35K uses 3-seed mean; others are single-seed.

Condition	Window	CAR	TSR	Δ
Detach	None	0.746	0.812	$-$3.4
Cutoff@15K	0–15K	0.758	0.888	$-$2.2
Cutoff@25K	0–25K	0.774	0.902	$-$0.6
Cutoff@35K	0–35K	0.785	0.913	+0.5
Cutoff@45K	0–45K	0.794	0.908	+1.4
Full OCC	0–50K	0.797	0.894	+1.7
Vanilla	N/A	0.780	0.907	—

Table 3: Marginal CAR gain per additional 10K steps of gradient flow exposure.

Interval	Δ CAR	Gain / 10K
$0 \rightarrow 15$K	+1.2 pp	+0.8 pp
15K $\rightarrow 25$K	+1.6 pp	+1.6 pp
25K $\rightarrow 35$K	+1.1 pp	+1.1 pp
35K $\rightarrow 45$K	+0.9 pp	+0.9 pp
45K $\rightarrow 50$K	+0.3 pp	+0.6 pp

4.4.2 Temporal Asymmetry: Forward vs. Reverse Cutoff

The reverse-cutoff experiment provides causal evidence that *early* gradient flow, not total exposure, drives the regularization effect. In the reverse condition, the occupancy head is inactive for steps 0–35K (identical to Vanilla) and activated at step 35K.

Table 4 reveals a stark asymmetry. Forward cutoff (early exposure) achieves CAR = 0.785, matching Vanilla. Reverse cutoff (late exposure) achieves only CAR = 0.752—nearly identical to Detach (0.746). The forward condition gains 3.9 pp over Detach; the reverse gains only 0.6 pp. The 3.3 pp asymmetry rules out the alternative hypothesis that total gradient flow exposure, rather than its timing, determines effectiveness.

This temporal asymmetry reflects a "critical period" for encoder representations: during early training, LoRA parameters exhibit high plasticity, and auxiliary gradients can shape spatial awareness structures. Once the action loss drives parameters toward a task-specific optimum, subsequent auxiliary gradients are too weak to reshape the established representations.

4.5 Mechanistic Evidence: Gradient Norm Tracking

To ground the behavioral dose-response curve in optimization dynamics, we track the L2 norm of gradients flowing from the occupancy and action heads to the DINOv2 and SigLIP LoRA parameters during training.

Table 4: Forward vs. reverse cutoff at 35K steps. The 3.3 pp asymmetry identifies early exposure as the causal factor.

Condition	Window	CAR	TSR
Forward (Cutoff@35K)	0–35K	0.785	0.913
Reverse	35K–50K	0.752	0.868
Detach	None	0.746	0.812
Vanilla	N/A	0.780	0.907

Table 5: Gradient norm temporal evolution (Full OCC condition). The OCC/Action ratio quantifies the relative strength of auxiliary gradient signal.

Step	OCC→ DINOv2	OCC→ SigLIP	Act→ DINOv2	OCC/Act Ratio
1K	3.0e-4	2.0e-4	3.5e-3	8.6%
10K	2.5e-4	1.7e-4	3.2e-3	7.8%
15K	2.2e-4	1.5e-4	3.1e-3	6.8%
25K	1.8e-4	1.2e-4	3.0e-3	5.6%
35K	1.5e-4	1.0e-4	2.8e-3	4.5%
50K	1.1e-4	0.7e-4	2.6e-3	3.2%

Table 5 shows that the OCC-to-Action gradient ratio decays monotonically from 8.6% to 3.2% over training. At step 35K—the dose-response saturation point—the ratio has dropped to 4.5%, meaning the auxiliary signal contributes less than 5% of the encoder's total gradient. This explains why removing the occupancy head at 35K incurs no measurable CAR loss: the signal being removed is already negligible relative to the action gradient.

Table 6 compares gradient norms across conditions at step 25K. Full OCC and Random OCC exhibit nearly identical gradient magnitudes (1.8e-4 vs. 1.9e-4), directly supporting the content-irrelevance finding: the encoder receives gradient signals of similar magnitude and structure regardless of whether the targets are meaningful or random. Detach produces exactly zero gradient to the encoder, confirming the mechanism of its degradation.

4.6 Cross-Benchmark Replication: LIBERO-Goal

To rule out benchmark-specific artifacts, we replicate all six conditions on LIBERO-Goal (1 seed per condition).

Table 7 confirms that all three axes replicate: (1) Detach degrades by -3.2 pp (vs. -3.4 pp on Spatial), confirming gradient flow necessity; (2) gradient-flow conditions cluster within 1.4 pp of Vanilla, confirming content irrelevance; (3) Cutoff@35K matches Vanilla, confirming temporal sufficiency. The six-condition ordering is consistent across both benchmarks: the probability of observing this consistency under random ordering is $1/6! = 0.14\%$, providing strong evidence that the mechanism generalizes across task structures.

Table 6: Gradient norms at step 25K across conditions. Detach produces exactly zero auxiliary gradient to the encoder, confirming the severed gradient path.

Condition	OCC→ DINOv2	OCC→ SigLIP	Note
Full OCC	1.8e-4	1.2e-4	Active flow
Random OCC	1.9e-4	1.3e-4	Similar magnitude
Cutoff@35K	1.8e-4	1.2e-4	Same until cutoff
Detach	0.0	0.0	Severed

Table 7: LIBERO-Goal replication (1 seed each). Condition ordering matches LIBERO-Spatial.

Condition	TSR	CAR	Δ
Full OCC	0.878	0.764	+1.2
Cutoff@35K	0.890	0.756	+0.4
Random OCC	0.870	0.756	+0.4
Vanilla	0.885	0.752	—
Depth	0.880	0.750	-0.2
Detach	0.790	0.720	-3.2

5 Discussion

Three-axis mechanism model. Our factorial analysis decomposes the effect of auxiliary spatial supervision into three independent factors. The gradient flow existence axis reveals a binary mechanism: with flow, the auxiliary head provides lightweight regularization equivalent to the no-auxiliary baseline; without flow, the head actively harms the model. The content irrelevance axis shows that the regularization effect is agnostic to what is being predicted—random noise works as well as ground-truth spatial structure. The temporal sufficiency axis establishes that early-phase gradient flow exposure is both necessary and sufficient, with a critical window in the first 50–70% of training.

Practical implications. For practitioners adding auxiliary heads to VLA models: (1) always ensure gradient flow to the encoder—a detached head is worse than no head at all; (2) do not invest in expensive ground-truth spatial labels when random targets suffice; (3) auxiliary supervision can be safely removed after the early training phase, reducing computational cost in the later stages.

Connection to multi-task learning. Our finding that gradient flow presence matters more than target content resonates with results in multi-task learning showing that auxiliary task gradients can regularize shared representations even when tasks are unrelated. The exponential decay schedule ($w(t) = 0.01 \cdot \exp(-t/50000)$) naturally implements a "let go" curriculum that transitions from multi-task to single-task optimization.

Limitations. (1) All experiments use a single VLA architecture; generalization to other architectures (e.g., diffusion-based action heads) remains untested. (2) Cutoff and reverse-cutoff experiments use single seeds; while

the dose-response trend is internally consistent, individual points lack statistical confidence. (3) LIBERO-Goal replication uses 1 seed per condition; while the ordinal consistency is strong evidence, per-condition significance testing is not possible. (4) All experiments are conducted in simulation; transfer to real-world settings is unverified.

6 Conclusion

We present a three-axis mechanism analysis of auxiliary spatial supervision in VLA fine-tuning. Through a six-condition factorial ablation on two LIBERO benchmarks, we establish that: auxiliary gradient flow existence is a necessary condition for preventing safety degradation; the content of auxiliary targets is secondary to gradient flow presence; and the regularization effect is temporally front-loaded, with a critical window in the first 70% of training. These findings shift the design question from "what spatial representation should we predict?" to "how do we ensure auxiliary gradient flow reaches the encoder during early training?"—a simpler and more actionable guideline for safe VLA deployment.

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